



MAT2041 · Assignment 4

Instructions: Please ensure that you complete this assignment independently. If your solution appears to be identical or highly similar to others or generated using AI, you may be required to meet with the TA to verify your understanding. Cases of academic dishonesty may result in penalties, including a zero score for the assignment or other actions by the disciplinary committee. If you have any questions or need assistance, feel free to reach out to the course staff for guidance. We encourage you to engage in the learning process with integrity and responsibility.

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Rules

- **English:** Answer the questions in **English**. Otherwise, you'll lose half of the points.
- **Electronic submission:** Turn in solutions electronically via Blackboard. Be sure to submit your homework as **a single readable file**.
- **Collaboration policy:** Collaboration is allowed for all problems, but please list all the people with whom you discussed. Crediting help from other classmates will not take away any credit from you. Notably, only insightful discussions are allowed. **Directly sharing the solutions is prohibited**. The details of the collaboration policy for this course are available in the Resources tab on Piazza.
- **Questions on homework.** Start early and come to TA office hours with your questions on the assignments.

Total: 80 points, 6 problems

1. (Short-Answer Questions)

(4 × 6 = 24 points)

Judge whether each statement is true or false.

(a) The vectors $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ form a basis of $\mathbb{R}^2$.(b) The vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ form a basis of $\mathbb{R}^3$.

(c) If the columns of a matrix are linearly dependent, then the rows of this matrix must also be linearly dependent.

(d) Let us consider the plane $V_p = \{(x, y, z)^T \in \mathbb{R}^3 \mid 3x + 2y - z = 0\}$. Then the dimension of the intersection of V_p with the plane $\{(x, y, z)^T \in \mathbb{R}^3 \mid x = 0\}$ is equal to 1.

(e) There exists a linear system of equations whose solution set is exactly the set of integers.

(f) Suppose A is an n by n matrix, then its columns are linearly independent iff $Ax=b$ has at least one solution for any b in $\mathbb{R}^n$.**Remark:** For judgment questions, you are suggested to write the solution as follows. “ (a) True. Reason: ” or “ (a) False. Counter-example: ”**Grading scheme:** 1 point for correct judgment, and 2 points for reasons or counter-examples.

2. Check whether the following vectors are linearly independent or dependent. If they are linearly dependent, express one of these vectors as a linear combination of the others. (2 × 5 = 10 points)

(a)

$$\alpha_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 4 \\ 2 \\ 1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}$$

(b)

$$\alpha_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 4 \\ 3 \\ -2 \\ 2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 4 \\ 6 \\ -1 \\ 4 \end{bmatrix}, \alpha_4 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

3. Calculate the rank of the following matrix.

(10 points)

$$A = \begin{bmatrix} 3 & 1 & 2 & 4 \\ 1 & 2 & -1 & 3 \\ -1 & 2 & 1 & 2 \\ 0 & 4 & -3 & 1 \end{bmatrix}$$

4. (Rank)

(6 + 6 = 12 points)

You can use the following result in this problem: Elementary row operations do not change the rank of a matrix.

(a) Let $A \in \mathbb{R}^{n \times k}$ and $D \in \mathbb{R}^{m \times p}$, prove that $\text{rank}\begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} = \text{rank}(A) + \text{rank}(D)$

(b) Let $A \in \mathbb{R}^{n \times m}$, $B \in \mathbb{R}^{m \times p}$. Prove: $\text{rank}(AB) \leq \text{rank}(A)$.

[Hint: Consider the row spaces or column spaces.]

5. (Subspace and Basis)

(4 + 2 + 4 + 4 = 14 points)

Let $A \in \mathbb{R}^{n \times n}$, $V = \{B \in \mathbb{R}^{n \times n} | AB = BA\}$.

(a) Prove that V is a subspace of $\mathbb{R}^{n \times n}$.

(b) Give a simple expression for V when $A = I_n$.

(c) Provide a basis for V when $n = 3$, $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.

(d) Let $P_2 = \{a_0 + a_1x + a_2x^2 \mid a_i \in \mathbb{R}\}$, define three polynomials as follows:

$$g_0(x) = 1, \quad g_1(x) = 1 + x, \quad g_2(x) = 1 + x + x^2.$$

Show that the set $B' = \{g_0, g_1, g_2\}$ is a basis of P_2 .

6. (Linear independence)

(3 + 3 + 4 = 10 points)

Let $v_1, \dots, v_n \in \mathbb{R}^m$, prove the following:

(a) if $v_1, \dots, v_n$ are dependent, then for any $A \in \mathbb{R}^{p \times m}$, $Av_1, \dots, Av_n$ are dependent.

(b) if $v_1, \dots, v_n$ are independent, and $A \in \mathbb{R}^{m \times m}$ is invertible, then $Av_1, \dots, Av_n$ are independent.

(c) if $Av_1, \dots, Av_n$ are independent, then $v_1, \dots, v_n$ are independent.